

Here's to 2025's epic science!

As someone who's spent years analysing consumer tech, I've decided to turn my gaze skyward – and inward – for a change. As we approach 2025, I find myself more excited about the scientific breakthroughs that promise to redefine not just our tech shelves, but our very understanding of life and the universe. Make no mistakes, this coming year isn't just another lap around the sun – it's shaping up to be a milestone in human scientific achievement. From revolutionary strides in genomics to ambitious space missions, 2025 is the year science takes centre stage. Here are a few things I'm excited about...

I've always been fascinated by the idea of editing genes – tweaking the very blueprint of life. Enter CRISPR, the gene-editing technology that's been making headlines for its potential to cure genetic diseases. But like any tool, its effectiveness depends on precision, and that's where AI comes into play.

Imagine AI algorithms sifting through vast genomic data to pinpoint the exact spot in our DNA that needs a tweak. It's like having a GPS for gene editing. Scientists are using AI to design guide RNAs that improve the accuracy of CRISPR, reducing off-target effects and unintended consequences.

The US FDA has projected that by 2025, it will be approving 10 to 20 cell and gene therapy products annually, reflecting the rapid progress in this field. What does this mean for us? Well, we're looking at the possibility of curing genetic disorders that were once considered a life sentence. With advancements in gene therapies like CRISPR-Cas9, doctors can edit genes within their own cells to treat diseases. Take, for example, the recent approval of a CRISPR-based therapy for sickle cell disease. By editing patients' stem cells, we can enable the production of healthy red blood cells, reducing the need for transfusions. Diseases like cystic fibrosis or certain forms of blindness could become relics of the past. It's a thrilling prospect that goes beyond any app update or gadget release, if you ask me.

Blame it on too many sci-fi movies as a kid (or even now), but I've always been a bit of a space nerd. And 2025 is all about some big ticket items toward exploring our solar system and beyond.



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Executive Editor



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NASA's Artemis II is scheduled to be the first crewed mission in the Artemis program, targeting a launch in 2025. This mission will carry astronauts on a lunar flyby, testing the Space Launch System (SLS) rocket and the Orion spacecraft's capabilities with crew aboard. The primary objective is to validate systems for subsequent lunar landings, marking a significant step toward sustainable lunar exploration. Closer to home, the NASA-ISRO Synthetic Aperture Radar (NISAR) mission will launch an Earth-observing satellite to monitor changes in our planet's ecosystems. With climate change being the defining issue of our time, this mission couldn't be more critical. And let's not forget ESA's Space Rider test flight and ISRO's collaboration with Japan on the Lunar Polar Exploration Mission. These endeavors aren't just about national pride, but about humanity pushing the boundaries of what's possible and inspiring the next generation of space nerds and dreamers.

You might be wondering why a tech columnist is waxing lyrical about genes and space missions. Simply put, without these scientific breakthroughs, our beloved gadgets just wouldn't exist. Every smartphone, every piece of software, stands on the shoulders of innumerable scientific discoveries. The advancements happening in labs and space agencies around the world have the potential to solve real problems, ultimately trickling down to everyday tech you and I will end up using soon enough.

Sure, I might have given up on making New Year's resolutions – after failing to lose weight and be fluent in a foreign language for the third straight year, who wouldn't? But if there's one thing I'm resolved to do, it's to stay curious and keep sharing these incredible stories of human scientific achievement with you.

So here's to a year where we celebrate the scientists, the researchers, and the dreamers who are turning what was once science fiction into science fact. And who knows? Maybe this will be the year I finally stick to a resolution (don't hold your breath). I'd love to hear your thoughts and, dare I say, your own resolutions – scientific or otherwise. Feel free to drop me a line at jayesh@digit.in. Stay curious, stay hopeful, and here's to a remarkable 2025. 🚀